An Introduction To Type-E Chopper
&
Multiphase Chopper

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⊞ **Type-E Chopper:** The load current as well as load voltage are either positive or negative which provide the four quadrants of operation (shown in Fig. 1).
Also known as "Four-Quadrant Chopper" or "Full-Bridge DC-DC Converter".
It consists of four (power) semiconductor switches along with an anti-parallel diode across the each switch.
NOTE: for operation in the third and fourth quadrants, the direction of the load emf E must be reversed internally.

Three distinct applications of the full-bridge switch-mode converter are as follows:

- DC-Motor Drive
- DC-to-AC (sine-wave)
- DC-to-AC (high intermediate frequency)

Here the load is a DC motor. We assume that the DC voltage source is ideal. Two switches of same leg must not on simultaneously, otherwise short circuit across the source will be occurred.

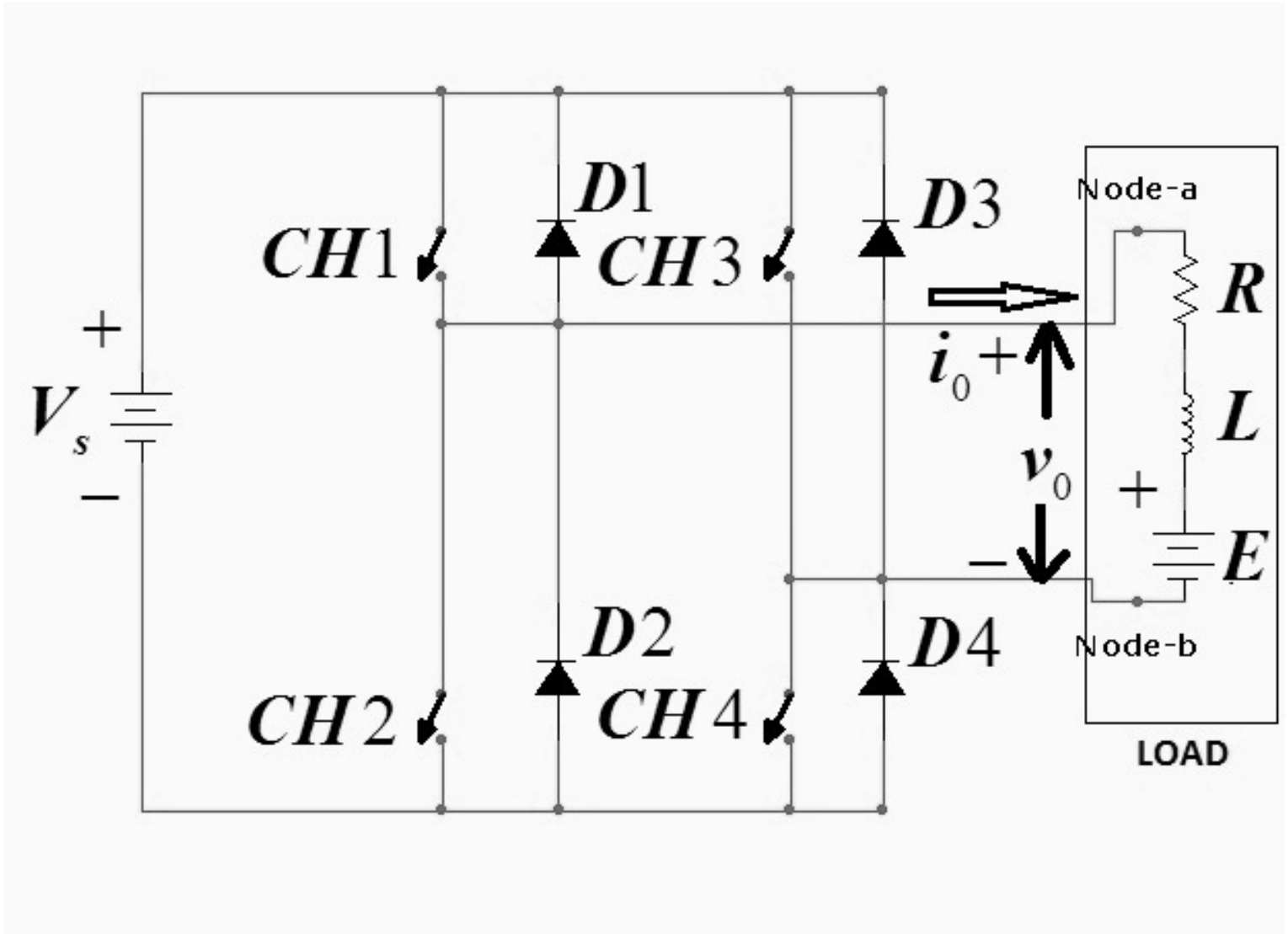


Figure 1: Circuit Diagram of a Type-E Chopper

★ First Quadrant:

Figure 2: Type-E Chopper: First Quadrant

★ Second Quadrant:

Figure 3: Type-E Chopper: Second Quadrant

★ Third Quadrant:

Figure 4: Type-E Chopper: Third Quadrant

★ Fourth Quadrant:

Figure 5: Type-E Chopper: Fourth Quadrant

★ Conclusion of Type-E Chopper:

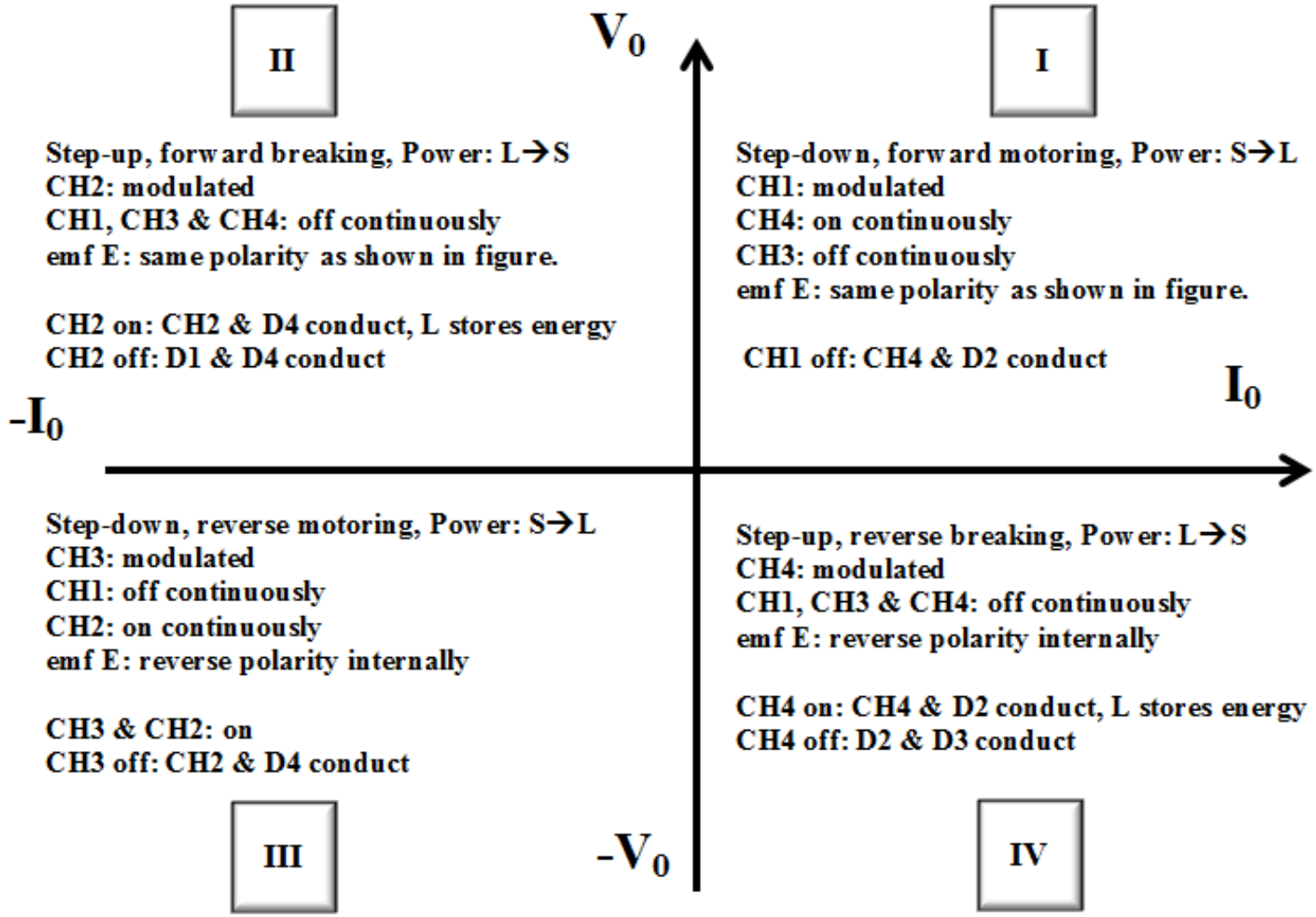


Figure 6: Conduction and polarity of devices of a Type-E Chopper

⊞ **Multiphase Chopper:** A multiphase chopper consists of tow or more choppers connected in parallel. A multiphase (two-chopper) is shown fig. 7

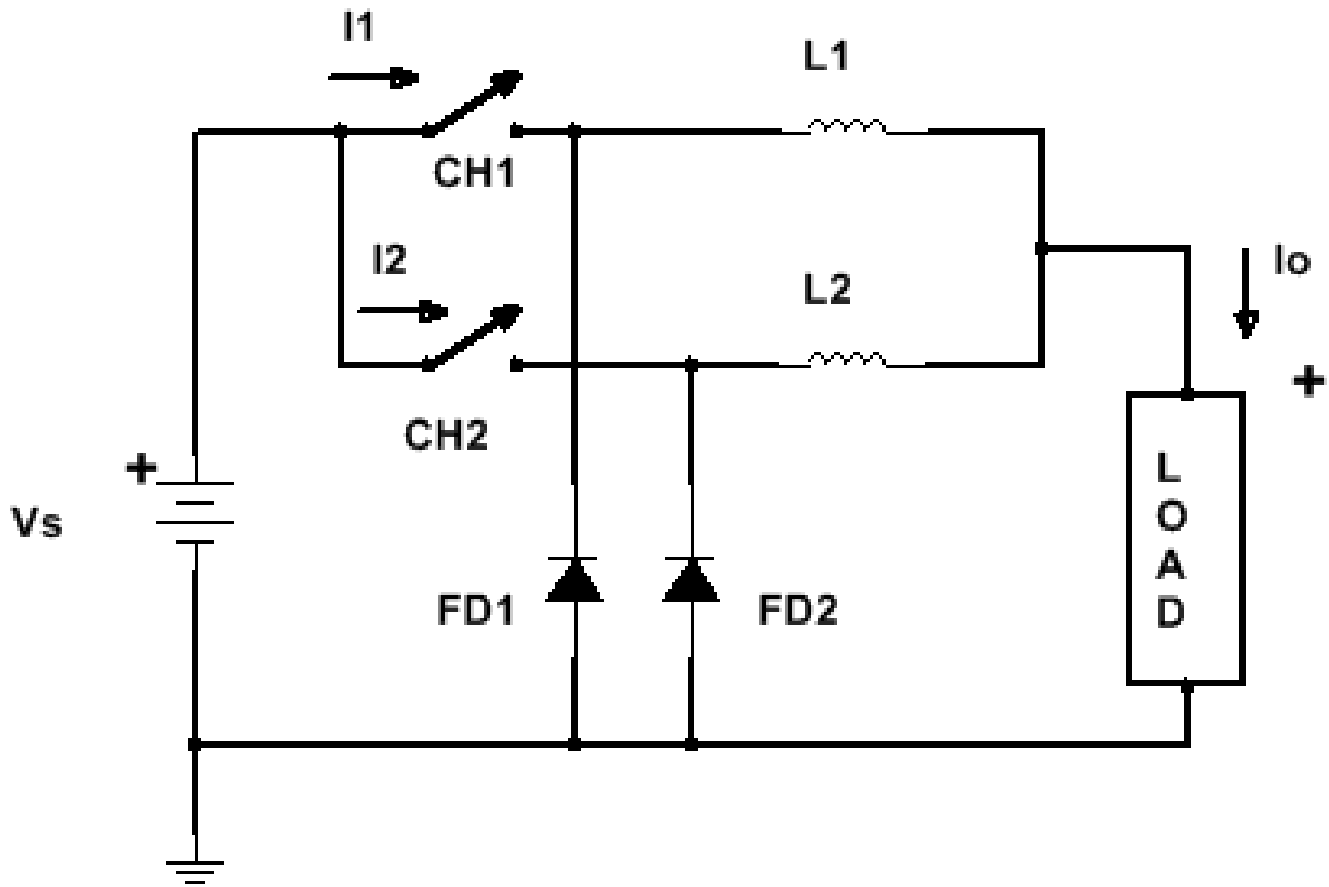


Figure 7: Circuit Diagram of a Multiphase (two-chopper) Chopper

It may be operated as follows:

- In-phase operation mode
- Phase-shifted operation mode

The input currents are shown in fig. 8 for in-phase mode operation

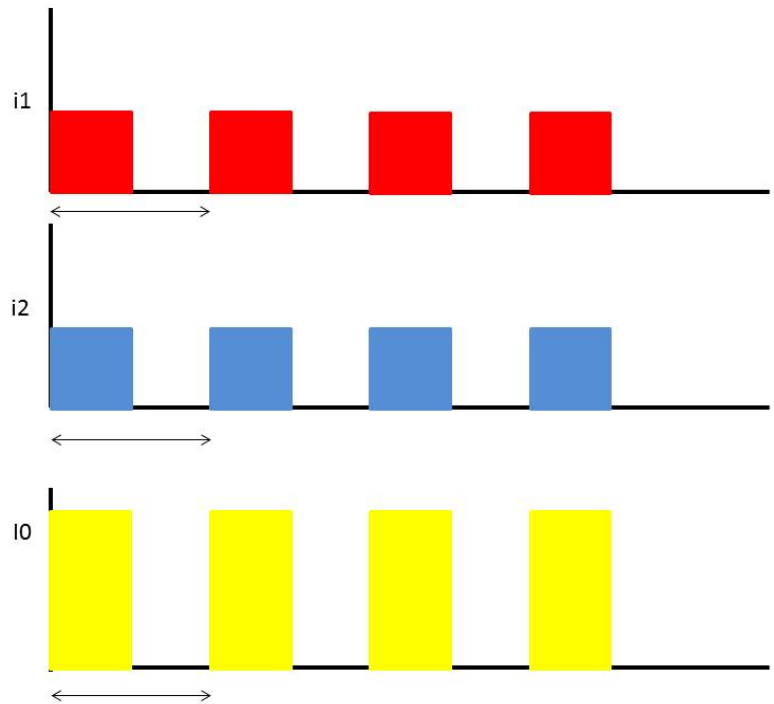


Figure 8: In-Phase Operation

The input currents are shown in fig. 9 for phase-shifted mode operation and duty ratio is gradually changing.

Figure 9: Phase-Shifted Mode Operation

★ Conclusion of Multiphase Chopper:

- **In-phase operation mode**:
  The input currents are added. Hence the output average load current (Io) will be double. Therefore equivalent to single-chopper operation.
- **Phase-shifted operation mode**:
  Input currents frequency becomes double , hence, filter size and cost will be less.
  The current ripple amplitude becomes half as that of in-phase operation.

Thank You...